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Plasma placental RNA allelic ratio permits noninvasive prenatal chromosomal aneuploidy detection

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Current methods for prenatal diagnosis of chromosomal aneuploidies involve the invasive sampling of fetal materials using procedures such as amniocentesis or chorionic villus sampling and constitute a finite risk to the fetus. Here, we outline a strategy for fetal chromosome dosage assessment that can be performed noninvasively through analysis of placental expressed mRNA in maternal plasma. We achieved noninvasive prenatal diagnosis of fetal trisomy 21 by determining the ratio between alleles of a single-nucleotide polymorphism (SNP) in *PLAC4* mRNA, which is transcribed from chromosome 21 and expressed by the placenta, in maternal plasma. *PLAC4* mRNA in maternal plasma was fetal derived and cleared after delivery. The allelic ratios in maternal plasma correlated with those in the placenta. Fetal trisomy 21 was detected noninvasively in 90% of cases and excluded in 96.5% of controls.

Current invasive prenatal diagnostic methods for fetal chromosomal aneuploidies, such as amniocentesis or chorionic villus sampling, pose a finite risk to the fetus[1]. Noninvasive screening methods, including nuchal translucency and maternal serum biochemical screening[2,3], typically detect the associated epiphenomena instead of the core genetic abnormalities of chromosome dosage imbalance and are not diagnostic in nature. For many years, research in noninvasive testing has focused on the isolation and analysis of the rare fetal nucleated cells that have entered into the maternal circulation[4–6]. Robust diagnostic testing, however, has not yet been achieved[7].

We previously reported the presence of cell-free fetal DNA in the plasma of pregnant women[8], which constitutes a mean of 3-6% of total maternal plasma DNA[9]. As a result of the overwhelming maternal DNA background in maternal plasma[9], direct assessment of fetal chromosomal copy number is not yet feasible[10]. An elevation

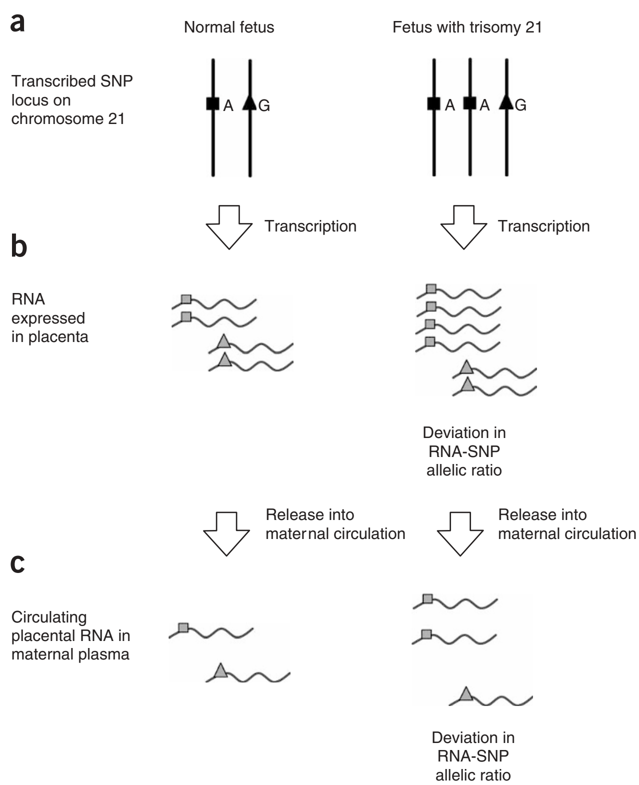

Figure 1 Schematic of the RNA-SNP allelic ratio strategy for noninvasive detection of fetal trisomy 21 through maternal plasma analysis by the relative quantification of alleles of placenta-expressed transcripts located on chromosome 21. (**a**) Normal and trisomy 21 fetuses heterozygous at the transcribed SNP locus of *PLAC4*. The fetus with trisomy 21 has an extra copy of the gene. "A" and "G" denote the respective alleles of the transcribed SNP. (**b**) The gene is expressed in placental tissues and the resultant transcripts are allelic with respect to the SNP. The ratio of the two RNA alleles in a trisomy 21 placenta is expected to deviate from that of a normal placenta, as a result of the expression of an extra copy of the gene. (**c**) The RNA transcripts are released into maternal blood and their relative abundance is reflective of that in the placenta. Thus, the plasma ratio of the two RNA alleles in a trisomy-21 pregnancy is expected to deviate from that of a normal pregnancy.

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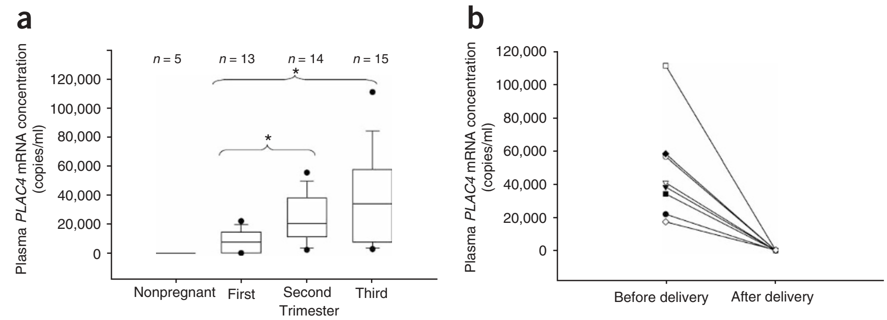

Figure 2 Detection of *PLAC4* mRNA in maternal plasma. (**a**) Asterisks indicate the groups with significant differences in plasma *PLAC4* mRNA concentrations ($P < 0.05$, Kruskal-Wallis test followed by pairwise comparison using the Student-Neuman-Keuls test). (**b**) Clearance of *PLAC4* mRNA from maternal plasma 24 h after delivery.

in circulating fetal DNA concentration has been reported in trisomy 21 pregnancies[11,12], but the large overlap in concentrations between normal and aneuploid pregnancies make the diagnostic impact of such an approach rather modest[13].

To achieve diagnosis of fetal chromosomal aneuploidy from maternal plasma, we reasoned that it would be necessary to target nucleic acid molecules, which were virtually fetal-specific (that is, free from maternal background interference), and then to derive chromosome dosage information from such molecules. We explored this approach by targeting fetal RNA of placental origin[14,15]. To provide chromosome dosage information, the circulating placental mRNA has to be transcribed from the chromosome of interest, for example, chromosome 21 for the diagnosis of trisomy 21 (ref. 16). We hypothesized that chromosome dosage could be determined by assessing the ratio between alleles of a SNP in the placenta-expressed RNA, the RNA-SNP allelic ratio, in a heterozygous fetus (**Fig. 1**). For example, for a SNP with two alleles, A and B, and assuming that the ratio of their expression levels in the placenta of a heterozygous euploid fetus is 1:1, we hypothesized that the corresponding ratio in the placenta of a heterozygous trisomic fetus would be 1:2 or 2:1. We further hypothesized that the RNA-SNP allelic ratio of the placental mRNA released into the plasma of a pregnant woman would reflect the allelic ratio in the placenta itself. By determining this ratio in maternal plasma, the number of copies of chromosome 21 that the fetus possesses can thereby be deduced noninvasively.

RESULTS

Placental expressed mRNA for maternal plasma detection

We first identified a gene on chromosome 21, which was expressed in the placenta and not expressed by the maternal cells that contributed to the background maternal-derived RNA in maternal plasma. We had previously argued that the maternal hematopoietic cells were likely to be the major contributor to the background nucleic acids[17–19]. By comparing the gene-expression patterns of the placenta and maternal blood cells, we previously developed a microarray-based strategy for identifying placental mRNA that would be detectable in maternal plasma[17]. Here, we mined this data set for placental mRNA species transcribed from chromosome 21. The identified transcripts are listed in **Supplementary Table 1** online. *PLAC4* (ref. 20) showed the highest absolute level of expression in the placenta and the largest relative difference in expression levels between the placenta and maternal blood cells. We next confirmed that *PLAC4* mRNA was detected in the plasma of women in all three trimesters of pregnancy (**Fig. 2a**), but not in the plasma of nonpregnant individuals (**Fig. 2a**). We observed clearance of *PLAC4* mRNA from maternal plasma within 24 h after delivery (**Fig. 2b**), thus showing its specificity to pregnancy.

We found four polymorphic SNPs in the transcribed regions of *PLAC4* (**Supplementary Table 2** online). We genotyped the maternal buffy coat and placental tissue DNA from 119 pairs of pregnant women and their fetuses for the most polymorphic SNP, rs8130833. To confirm the fetal origin of *PLAC4* mRNA molecules in maternal plasma, we compared the *PLAC4* SNP genotype in the placental and maternal plasma RNA samples of pregnancies in which the mother and fetus harbored different genotypes at the SNP. Eleven of the 119 pregnancies involved homozygous fetuses and heterozygous mothers for the SNP, or vice versa (**Supplementary Table 3** online). In all 11 cases, we detected identical *PLAC4* mRNA SNP genotypes in both the placental tissues and maternal plasma, consistent with those of the placental DNA samples (**Supplementary Fig. 1** and **Supplementary Table 3** online). Among the homozygous fetuses, the maternal-specific *PLAC4* mRNA allele was not detected in the placental tissues or maternal plasma. These results confirmed that *PLAC4* mRNA in maternal plasma was derived from the fetus with no contribution from maternal tissue sources. Because of the lack of interference from background maternal molecules, the proposed strategy for fetal chromosome dosage assessment would be applicable to all cases in which the fetus was heterozygous for the targeted SNP, regardless of the maternal genotype. Among this cohort of

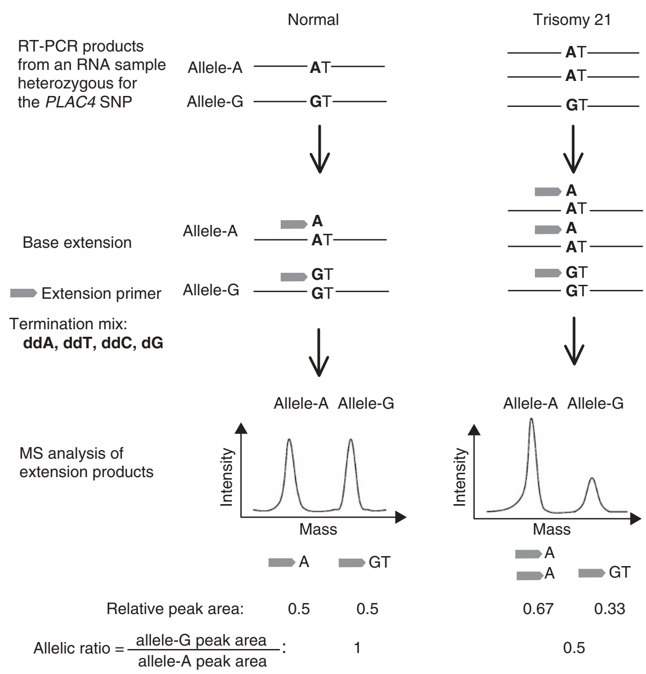

Figure 3 Schematic illustration of the analytical approach for RNA-SNP allelic ratio determination. The relative abundance of the two SNP alleles in heterozygous fetuses as determined by primer extension and mass spectrometry allows normal and trisomy 21 fetuses to be distinguished. MS, mass spectrometry.

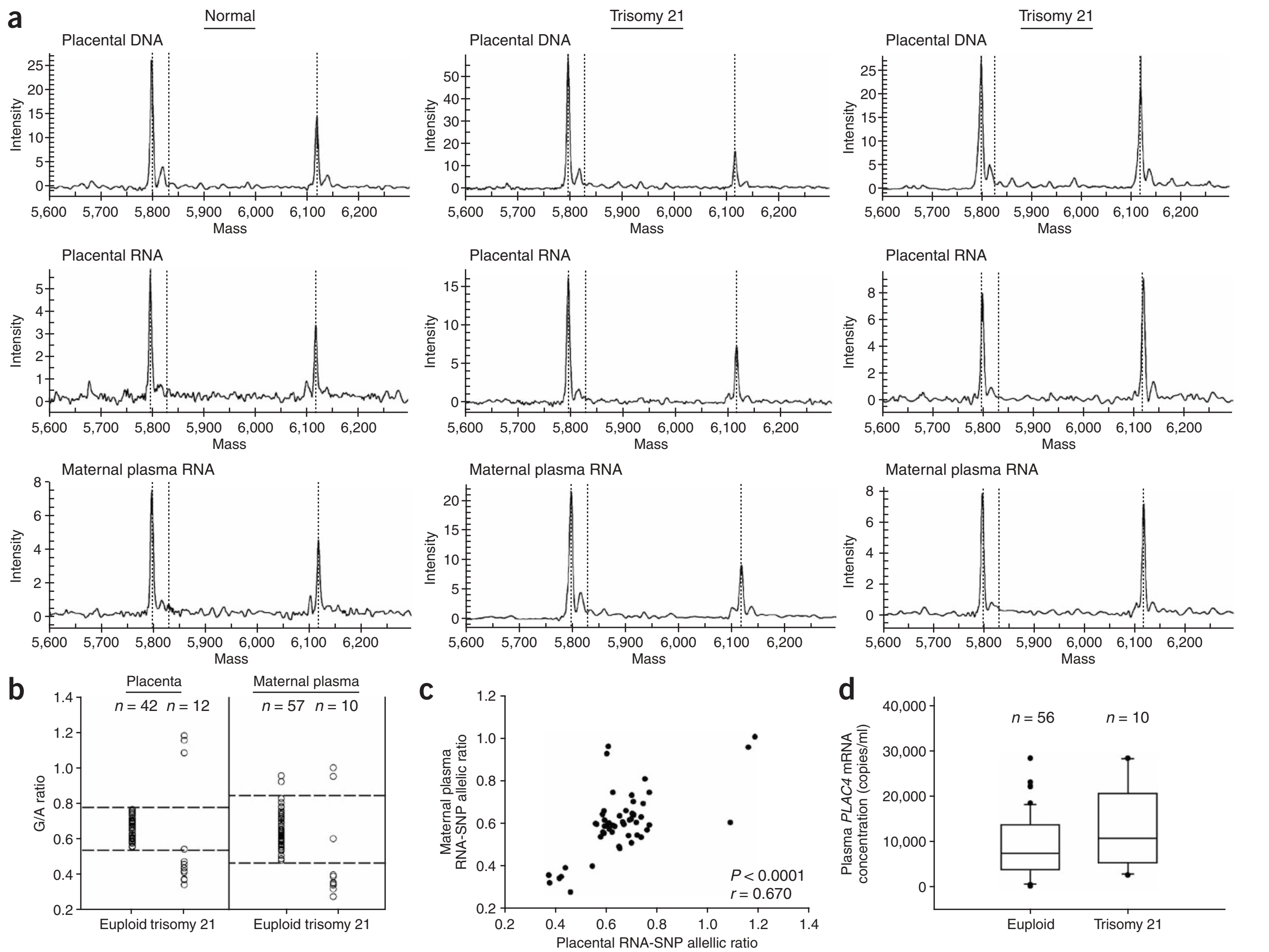

Figure 4 *PLAC4* mRNA for fetal trisomy-21 detection. (**a**) MS tracings of *PLAC4* allelic ratio determination in placental and maternal plasma samples. Left panels, samples from a pregnant woman carrying a euploid fetus. Middle panels, samples from a pregnant woman carrying a trisomy 21 fetus with an extra A allele. Right panels, samples from a pregnant woman carrying a trisomy 21 fetus with an extra G allele. The *x* axis depicts the molecular weights of the detected extension products (shown as sharp peaks), whereas the *y* axis depicts the intensity in arbitrary units. The extension product of the A allele is shown as the first peak, whereas the extension product of the G allele is shown as the second peak. (**b**) RNA-SNP allelic ratios of *PLAC4* in placentas and maternal plasma of euploid and trisomy-21 pregnancies. The ratios of the trisomy-21 samples segregated into two groups. Trisomy samples with an extra G allele showed a higher G/A ratio than the euploid group, whereas samples with an extra A allele showed a lower G/A ratio than the euploid group. The reference intervals, defined as the mean ± 1.96 s.d., are delineated by dashed lines. (**c**) Correlation between *PLAC4* RNA-SNP allelic ratios in plasma and placenta. (**d**) Absolute *PLAC4* mRNA concentrations in plasma of women carrying euploid and trisomy-21 fetuses.

119 pregnancies, 69 fetuses were shown to be heterozygous and thus their *PLAC4* RNA-SNP allelic ratio would be assessable.

PLAC4 RNA-SNP allelic ratio determination

We developed an approach (**Fig. 3**) for determining the allelic ratio of the SNP rs8130833 in *PLAC4* mRNA by primer extension followed by mass spectrometry detection of the extension products[21]. We carried out *PLAC4* RNA-SNP analysis on placental tissues of 42 euploid and 12 trisomy-21 fetuses heterozygous for this SNP (**Fig. 4a,b**). We calculated the RNA-SNP allelic ratios by dividing the relative amount of the higher-mass allele, allele G, by the relative amount of the lower-mass allele, allele A (**Fig. 3**). All trisomy-21 placental samples showed deviated RNA-SNP allelic ratios from that of the euploid group (**Fig. 4b**). We defined a reference interval as the mean RNA-SNP allelic ratio for euploid fetuses ± 1.96 s.d. Using this reference interval, which corresponded to 0.538 to 0.780, all but one trisomy-21 case showed allelic ratios that fell outside the interval, whereas all the euploid cases fell within the interval. These data translate to a sensitivity and specificity of 91.7% and 100%, respectively.

We next applied this approach to maternal plasma analysis (**Fig. 4a,b**) in 57 women carrying karyotypically normal fetuses (mean gestation, 13 weeks; range, 11.1 to 14 weeks) and ten women carrying trisomy-21 fetuses (mean gestation, 14.7 weeks; range, 12.4 to 20 weeks). The reference interval for maternal plasma corresponded to 0.461 to 0.841. Using this definition, two of the plasma samples from the euploid group had allelic ratios outside of the normal reference interval (**Fig. 4b**). All but one trisomy sample showed allelic ratios that deviated from the normal reference interval (**Fig. 4b**). These results translate to a sensitivity of 90% and a specificity of 96.5% for the detection of fetal trisomy 21, using just a single marker. Placental and plasma *PLAC4* RNA-SNP allelic ratios correlated with one another (**Fig. 4c**) (Pearson correlation, $P < 0.0001$, $r = 0.670$). Plasma RNA-SNP allelic ratio could thus be used as a noninvasive indicator of the placental allelic ratio and hence the fetal aneuploidy status.

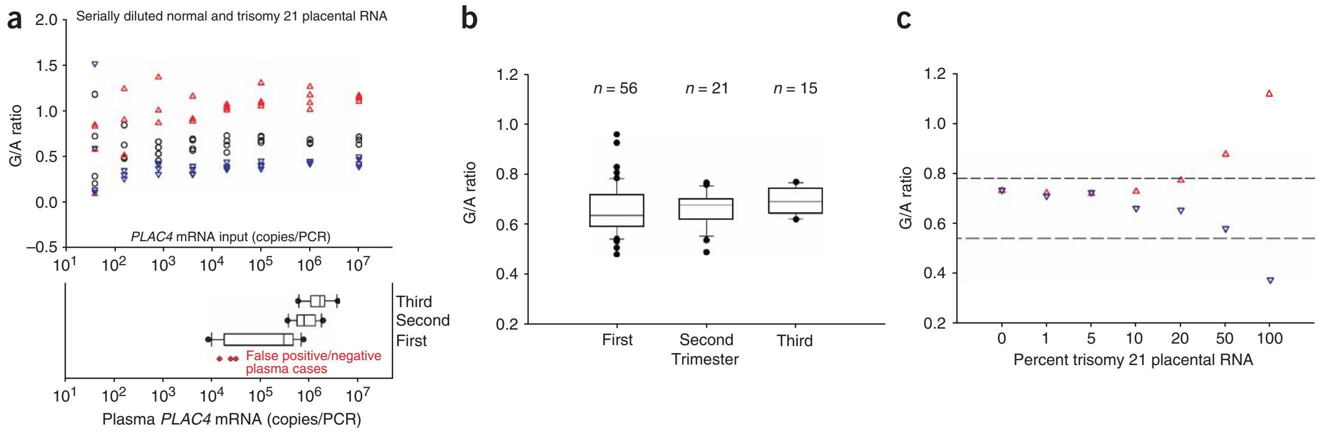

Figure 5 The relative insensitivity of RNA-SNP ratio analysis to gestational age and placental mosaicism. (**a**) Reproducibility of RNA-SNP ratio analysis in relationship to the amount of input *PLAC4* mRNA (top panel). The black open circles, blue inverted open triangles and red open triangles represent euploid fetuses and trisomy-21 fetuses with the AAG or AGG genotype for the *PLAC4* SNP rs8130833, respectively. The amount of input *PLAC4* mRNA is illustrated in reference to the concentration of circulating *PLAC4* mRNA in the three trimesters of pregnancy (bottom panel). Red filled diamonds mark the two false-positive cases and one false-negative case. (**b**) Maternal plasma RNA-SNP allelic ratios in women carrying euploid fetuses in the three trimesters of pregnancy. (**c**) RNA-SNP allelic ratio determination in placental RNA mixtures between a trisomy-21 fetus and a euploid fetus at varying proportions. Trisomy-21 fetus with the AAG (blue inverted open triangles) or AGG (red open triangles) genotype for the *PLAC4* SNP rs8130833. The reference intervals for placental tissue RNA-SNP allelic ratio are delineated by dashed lines.

Without the use of the RNA-SNP allelic ratio approach, the total maternal plasma concentration of *PLAC4* mRNA was not significantly different between the trisomy 21 and euploid cases (**Fig. 4d**) (Mann-Whitney test, $P = 0.165$).

Robustness of the RNA-SNP allelic ratio approach

We hypothesized that the abundance of *PLAC4* mRNA in maternal plasma might affect the accuracy of RNA-SNP allelic ratio determination. We serially diluted the total placental RNA from one euploid and two trisomy-21 fetuses to contain *PLAC4* mRNA concentrations ranging from 10^1 to 10^7 copies per PCR. We then determined the *PLAC4* RNA-SNP allelic ratios four times in each of the diluted aliquots of each of the three placentas. We achieved a clear separation of the trisomy-21 and euploid samples at *PLAC4* mRNA concentrations of 10^3 copies per reaction or above (**Fig. 5a**). We considered these data in reference to the concentrations of maternal plasma *PLAC4* mRNA in the three trimesters of pregnancy (**Fig. 5a**). The trisomy 21 and euploid cases could be distinguished by *PLAC4* RNA-SNP allelic ratio even at the *PLAC4* mRNA concentrations typical of that for first-trimester pregnancies. We further determined the maternal plasma *PLAC4* mRNA concentrations for the two false-positive and one false-negative cases identified by the RNA-SNP analysis. These three maternal plasma samples contained relatively lower concentrations of *PLAC4* mRNA and thus might have affected the accuracy of the determined RNA-SNP ratio.

We determined the *PLAC4* RNA-SNP allelic ratio in maternal plasma samples from the three trimesters of pregnancy involving euploid fetuses. The allelic ratios were not significantly different among the three trimesters (Kruskal-Wallis one-way analysis of variance, $P = 0.331$), indicating that such analysis was not affected by the gestational age (**Fig. 5b**).

Confined placental mosaicism, although uncommon, is a potential confounder for false-positive detection of fetal chromosomal aneuploidy[22,23]. We investigated if the RNA-SNP approach would produce false-positive results in this situation by simulating varying degrees of placental mosaicism. We mixed placental RNA from two trisomy-21 fetuses, respectively, with placental RNA from a euploid fetus at multiple *PLAC4* concentrations. We determined the *PLAC4* mRNA allelic ratio in each of the mixtures. The RNA-SNP allelic ratio did not deviate beyond the reference interval as defined above until the trisomy-21 placenta contributed 50% of the *PLAC4* RNA concentration (**Fig. 5c**). These data suggest that at the typical degrees of placental mosaicism encountered clinically[22], false-positive diagnosis of fetal trisomy 21 would be unlikely.

DISCUSSION

Here, we show that allelic ratio analysis of placental expressed mRNA transcripts in maternal plasma allows the noninvasive prenatal detection of trisomy 21. The key enabling elements of this strategy are described in the **Supplementary Note** online. The RNA-SNP ratio strategy allows the direct assessment of chromosome dosage using maternal plasma, which is fundamentally different from the serum biochemical markers currently used in screening for Down syndrome[2,3]. The diagnostic sensitivity and specificity of 90% and 96.5%, respectively, for maternal plasma RNA-SNP analysis using just a single marker in informative cases are already comparable to many multi-marker screening strategies[2,3]. There is a need to replicate these findings in a large-scale, multicenter setting. An additional advantage of the RNA-SNP allelic ratio strategy lies in its insensitivity to the gestational age at which sampling is performed, unlike the serum biochemical markers, thus simplifying the eventual clinical use of this strategy (**Fig. 5b**)[2]. This characteristic is expected as the core genetic defect of trisomy 21 has not changed over gestation.

The present method is applicable to a subset of the population as dictated by the heterozygosity rate of the studied SNP. According to the HapMap Project database (hapmap.org/), rs8130833 has a combined heterozygosity rate of 0.45 in the studied ethnic groups (**Supplementary Table 2**). We could further broaden the population coverage by extending the approach to other SNPs in the gene *PLAC4* and by testing other chromosome 21 transcripts identified by the microarray analysis. We have conducted a preliminary screen for SNPs within the transcribed regions of *PLAC4*, *COL6A1* and *COL6A2*

(**Supplementary Table 1**). We could readily achieve population coverage of 90% and 95% in Chinese and Caucasian fetuses, respectively, based on the use of a panel of seven SNPs in the three genes (data not shown). We envision that, if and when the RNA-SNP allelic ratio strategy is introduced into the clinical setting, each maternal plasma sample would be subjected to allelic detection for all SNPs in the panel. The allelic ratios would then be computed only for SNPs for which two alleles were detected, indicating fetal heterozygosity with respect to that SNP. As a result of the fetal specificity of the targeted transcripts, prior knowledge regarding the maternal or paternal genotypes would not be required.

The method could also be applied to the prenatal assessment of other chromosomal aneuploidies, for example, trisomy 18 and 13, through the mining of the relevant transcripts from the microarray data set. Outside prenatal diagnosis, the RNA-SNP allelic ratio method could also potentially be applied to the study of gene dosage variation of other cell types that are known to release RNA into the plasma, for example, cancer cells[24].

METHODS

Detailed methods are described in the **Supplementary Methods** online.

RNA-SNP allelic ratio. Determination of the RNA-SNP ratio involved the steps of reverse transcription of RNA, PCR amplification, base extension and analysis of the extension products by a mass spectrometer. Sequences of the oligonucleotides used in these reactions are shown in **Supplementary Table 4** online. PCR primers for SNP discovery from genomic DNA are listed in **Supplementary Table 5** online. Placental RNA (625 ng) or 48 µl plasma RNA was reversed transcribed in a reaction volume of 20 µl or 100 µl, respectively, using the ThermoScript reverse transcriptase (Invitrogen), according to the manufacturer's recommendation. Each reaction contained 50 nM gene-specific primer (Integrated DNA Technologies), 1 mM each of dATP, dTTP, dCTP and dGTP mix (Promega), 1× cDNA synthesis buffer, 5 mM DTT, 0.1 U/µl RNaseOUT and 0.0375 U/µl Thermoscript reverse transcriptase. After preincubating the mixture of RNA, primer and dNTP at 65 °C for 10 min, the reaction was carried out at 55 °C for 60 min followed by 85 °C at 5 min.

We then PCR amplified the resultant cDNA samples. We added all placental cDNA or plasma cDNA to a total PCR volume of 40 µl or 200 µl, respectively. Each reaction contained 0.6× HotStar *Taq* PCR buffer with 0.9 mM $MgCl_2$ included (Qiagen), 25 µM each of dATP, dGTP and dCTP, 50 µM of dUTP (Applied Biosystems), 200 nM each of forward and reverse primers (Integrated DNA Technologies) and 0.02 U/µl of HotStar *Taq* Polymerase (Qiagen). The PCR was initiated at 95 °C for 7 min, followed by denaturation at 95 °C for 40 s, annealing at 56 °C for 1 min, extension at 72 °C for 1 min for 55 cycles and a final incubation at 72 °C for 3 min. No product was observed when we subjected placental RNA samples to PCR without reverse transcription.

We used a Homogenous MassEXTEND (hME) assay (Sequenom) to determine the SNP allelic ratios (**Fig. 3**). The hME assay was based on the annealing of an extension primer adjacent to the polymorphic site. The primer was then extended through the polymorphic site for 1 or 2 bases, depending on the SNP allele. The molecular masses of the allele-specific extension products were then measured by the MassARRAY Compact (Sequenom), which is a matrix-assisted laser desorption and ionization time-of-flight (MALDI-TOF) mass spectrometer. We designed the assay such that the extension products for each SNP allele would show distinct, readily resolvable masses.

To perform the hME analysis, we first subjected PCR products to shrimp alkaline phosphatase (SAP) treatment to dephosphorylate any remaining dNTPs and prevent their incorporation in the subsequent primer extension assay. We added 0.34 µl hME buffer (Sequenom), 0.6 µl SAP (Sequenom) and 3.06 µl water to 25 µl PCR product. We incubated the reaction mixture at 37 °C for 40 min, followed by heat inactivation at 85 °C for 5 min.

For primer extension, we added 5 µl treated PCR product to 9 µl base extension reaction cocktail containing 342.6 nM extension primer (Integrated DNA Technologies), 0.51 U Thermosequenase (Sequenom) and 28.4 µM each of ddATP, ddCTP, ddTTP and dGTP (Sequenom). The thermal profile for the primer extension reaction was 94 °C for 2 min, followed by 100 cycles of 94 °C for 5 s, 52 °C for 5 s and 72 °C for 5 s.

Before MALDI-TOF analysis, the base extension products were cleaned up with the SpectroCLEAN (Sequenom) resin to remove salts that may interfere with mass spectometry analysis. We added 24 µl water and 12 mg resin to each base extension product and mixed the final mixtures in a rotator for 30 min. After centrifugation at 361*g* for 5 min, we dispensed approximately 10 nl reaction solution onto a 384-format SpectroCHIP (Sequenom) prespotted with a matrix of 3-hydroxypicolinic acid by using a SpectroPoint nanodispenser (Sequenom). A MassARRAY Analyzer Compact mass spectrometer (Bruker) was used for data acquisition. Data were automatically imported into SpectroTYPER (Sequenom) database for analysis.

For each heterozygous SNP, the relative amount of the two SNP alleles was represented by the ratio of the peak areas for the base extension products corresponding to the respective alleles and was determined using the SpectroTYPER software (Sequenom). The allelic ratio was then calculated by dividing the relative amount of allele G (the higher-mass allele, that is, the allele whereby the extension product shows a higher mass in the mass spectra) to the relative amount of allele A (the lower-mass allele). In theory, the G/A ratio should be 1 for a euploid fetus and 0.5 for a trisomy-21 fetus with genotype of AAG (**Fig. 3**). The observed ratios, however, are usually of a smaller value as a result of an inherent property of mass spectrometry whereby the signal intensity of the higher mass product (allele G in this case) is attenuated[25].

Note: Supplementary information is available on the Nature Medicine website.

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COMPETING INTERESTS STATEMENT
The authors declare competing financial interests (see the *Nature Medicine* website for details).

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